RICARDO ANTONIO AVALOS URIBE

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SUMMARY

Data Scientist with 4+ years of experience applying causal inference, statistical modeling, and machine learning to analyze large-scale datasets and quantify the impact of interventions. Proven expertise in designing experiments, conducting observational causal studies, and building end-to-end analytical pipelines to generate actionable insights. Strong background in translating complex analyses into decision-driven recommendations.

TECHNICAL SKILLS

Programming & Data: Python, SQL, PySpark, R, Stata
Causal Inference Methods: Experimental Design, A/B Testing, Difference-in-Differences, Causal ML, Network Analysis
Machine Learning: Random Forest, Gradient Boosting, Feature Engineering, Clustering
Data Engineering & Tools: Distributed Computing (Spark), Reproducible Research Workflows (Git), AI-assisted coding and analysis tools (LLMs for code generation, data analysis, and workflow automation)
Languages: English (Fluent), Spanish (Native)

EXPERIENCE

Research Analyst - University of Chicago Crime and Education Lab May. 2024 – Present
Data Project Associate - University of Chicago Crime and Education Lab Jan. 2023 - Apr 2024

- Evaluated causal inference studies to quantify the impact of medium to large scale criminal justice and education interventions, leveraging administrative datasets with millions of observations
- Conducted simulation-based power analyses to optimize experimental and quasi-experimental designs, ensuring statistical validity and reliable detection of treatment effects
- Developed automated end-to-end analysis workflows, including data preprocessing, hypothesis testing, treatment effect estimation, and automated generation of statistical outputs for decision-making and reporting (reduced turnaround time by 50%)
- Modeled network spillover effects using geo-referenced administrative data, quantifying indirect treatment effects and improving inference accuracy
- Led a team of three analysts, implementing Git-based version control workflows, task tracking systems, and structured project management processes to improve collaboration, ensure reproducibility, and accelerate delivery of analytical results to project stakeholders
- Presented causal inference findings to cross-functional stakeholders, including technical analysts and program leaders, translating complex statistical concepts such as spillover effects into recommendations to inform program strategy and intervention design

Research Assistant - MIDE Development Peru Mar. - Dec. 2022

- Applied experimental and quasi-experimental methods (RCTs, DiD, matching) to evaluate skills training and job intermediation programs targeting vulnerable populations
- Conceptualized experimental evaluation proposal using causal network analysis to increase technical certification among high school students in Costa Rica

Data Scientist Intern - Credit Bank of Peru (BCP) *Dec. 2019 – Dec. 2020*

- Developed machine learning models (Random Forest, Gradient Boosting) to predict credit risk for SME clients, improving predictive performance and supporting lending decisions
- Built customer segmentation models using unsupervised learning (K-means clustering), improving targeting and risk stratification
- Designed economic valuation framework to quantify the marginal value of alternative data sources, directly informing business negotiations and data acquisition strategy
- Collaborated with credit risk and product teams to translate model outputs into actionable business rules and risk management strategies

EDUCATION

Bs in Economics (top 5% ranked) | Pontifical Catholic University of Peru | Dec 2021
Relevant Coursework: Statistical Inference, Advanced Econometrics, Machine Learning, Evaluation of Social Policies